

11. (a) Iron reacts with halogens forming iron halides.

(i) Write a balanced equation for the formation of iron(III) bromide. [3]

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(ii) You are provided with solutions of iron(III) bromide and iron(II) iodide.

Describe a test to identify which is which.

Your answer should include the expected observation for both compounds. [3]

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(b) Bromine and fluorine react to produce a compound called bromine fluoride.

In a reaction 16.3 g of bromine reacted with 19.4 g of fluorine. Calculate the simplest formula for the bromine fluoride produced.

You **must** show your working. [3]

$$A_r(\text{Br}) = 80$$

$$A_r(\text{F}) = 19$$

Simplest formula

END OF PAPER

